WASP-1b

R-1077

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-1_181226` · created 2026-07-30T14:47:09+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458478.6422 +/- 0.0096 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.132 +/- 0.04
Transit depth (Rp/Rs)^2	1.74 +/- 1.06 [%]
Orbital Inclination (inc)	88.07 +/- 1.46
Airmass coefficient 1 (a1)	0.79 +/- 0.51
Airmass coefficient 2 (a2)	0.39 +/- 1.01
Scatter in the residuals of the lightcurve fit is	90.25 %
Best Comparison Star	#3 - [479, 53]
Optimal Aperture	2.77
Optimal Annulus	5.36
Transit Duration (day)	0.1471 +/- 0.0086

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.132 ± 0.04 vs 0.1036 (deviation 0.66σ)
Duration fit vs published	0.1471 d vs 0.1567 d (deviation 1.04σ)
Mid-time O–C	-23.0 min
Depth SNR	3.1
Residual scatter	90.25 %

Light curve

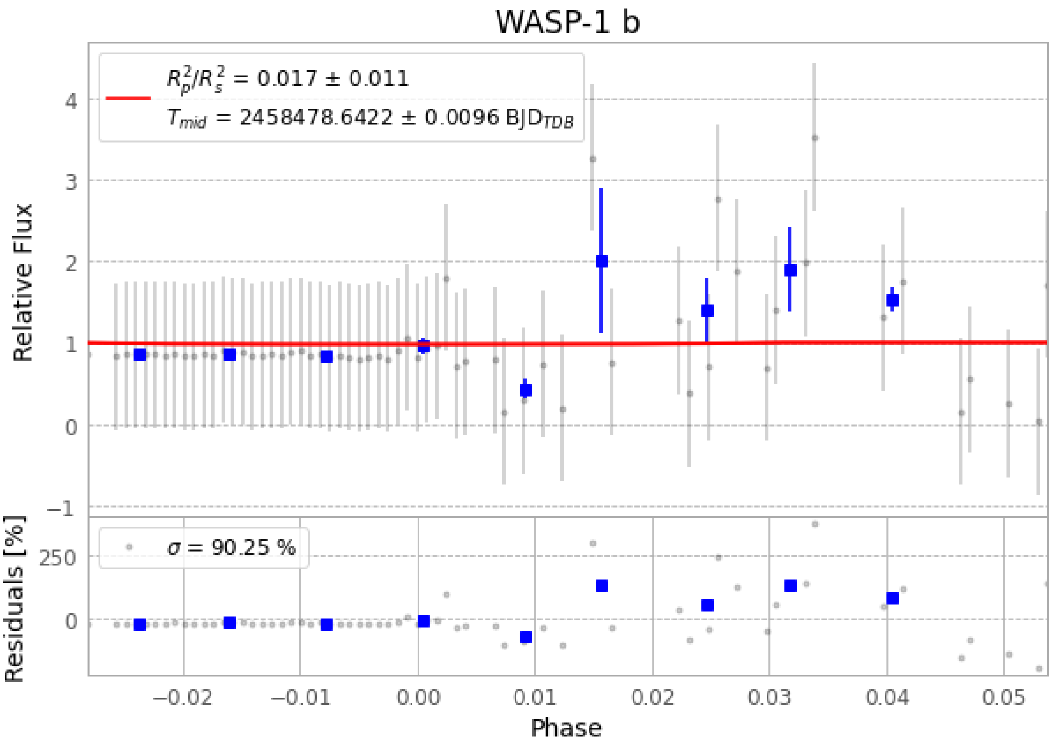

Figure 1 — FinalLightCurve_WASP-1 b_2018-12-25.png (SHA-256 5a7bc134031f04f5…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [332, 308], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 77aeeadfd04e6dd5…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

98 FITS frames (64 MB), `WASP-1181226013909.FITS` → `WASP-1181226063612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-181226.log (b066fb4150f5…), FinalLightCurve_WASP-1 b_2018-12-25.png (5a7bc134031f…), FinalLightCurve_WASP-1 b_2018-12-25.csv (90f8612855d2…)

Compute resources

Wall time 530.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)